YANG, Yan

Google Scholar · GitHub · LinkedIn

Degrees

Aug 2017 – May 2019 M.Sc. Computer Science, Georgetown University (Washington, DC)
* ***Supervisor:*** *Prof. Grace Hui Yang*

Sep 2002 – Jun 2006 B.Eng. Information Engineering, Southeast University (Nanjing, China)
* ***(2003)*** *Course Scholarship for Linear Algebra and Space Analytic Geometry* * *Admitted at age 16*

Other Education

(Aug 2024 - now) self-study:
• *The Principles of Quantum Mechanics*, P.A.M.Dirac (§1-20 / Ch 1-3; now on Ch 4)
• *Elements of Information Theory*, Tomas Cover (fully grasped Ch 1-8 and 14; touched on Ch 11, 13, and 15)
• MIT 18.102 *Introduction to Functional Analysis* (went over Lectures 1-8 and 14 Basic Hilbert Space Theory)
• *How Emotions Are Made: The Secret Life of the Brain*, Barrett, L.F. (Ch 1–9 of 11)

(Oct 2025) *Autumn School on fMRI*, @ IBRO & New York University, Abu Dhabi
• Exposure to experimental design and analysis workflows (Python-based), fMRI principles (HRF, preprocessing), and notebook-based modelling pipelines (e.g., GLM and second-order models).

(Sep 2023 - Nov 2024) Research stay, Computer Science, @ KAUST (Thuwal, Saudi Arabia)

(Jan 2021 – Aug 2022) Graduate research & coursework, Computer Science & Engineering, @ University of Nevada, Reno * ***Supervisors:*** *Prof. Mircea Nicolescu & Prof. George Bebis*

(Jan 2021 – May 2023) Studies on Coursera: *Probabilistic Graphical Model* (Part 2: 94/100; Part 1: 100/100), *Computational Neuroscience* (99.57/100), *Convolutional Neural Networks* (100/100), *Mathematical Biostatistics Boot Camp* (100/100)

Research Output

Publications & Preprints:

<Cognitive State Modelling>

• Yang, Y. **(2026)**. Modeling Limits in Phenomenological Contexts – a Statement on Luo et al.'s *Improving Searcher Struggle Detection via the Reversal Theory*. *Journal of Motivation, Emotion, and Personality* 14 (2026), pp. 1-5. doi.org/10.12689/jmep.2026.1401

• Yang, Y. **(2026)**. From Satiation to Allostasis: Toward a Resource-Based Framework for Reversal Triggers in Reversal Theory. *Zenodo*. https://zenodo.org/records/19437171

• Jiyun, L., Yang, Y., Nayak, V., & Yang, G.H. **(2024)**. Improving Searcher Struggle Detection via the Reversal Theory. *Discover Computing* 27, 51. Springer Nature. doi.org/10.1007/s10791-024-09492-z

<Rule-Informed ML for Modeling>

• Yang, Y. **(2026)**. Loss Landscape Diagnosis for Gradient-Based Gray-Scott System Inversion: Disentangling the Roles of PINN Components. *AI4Physics@ICML 2026 Workshop*; under major revision for resubmission to *TMLR*. https://arxiv.org/abs/2606.112

• Yang, Y. **(2026)**. EM-style Framework for Task-Driven Differentiable Synthesizer Optimization: Rationale, Design, and Backpropagation through Reaction–Diffusion System. *Zenodo*. https://zenodo.org/records/19397678

• Yang, Y., Bebis, G., & Nicolescu, M. **(2022)**. Two-Step Data Augmentation for Masked Face Detection and Recognition: Turning Fake Masks to Real. In Proceedings of *the 2nd International Conference on Image Processing and Vision Engineering – IMPROVE* (pp. 126-134). SciTePress. dx.doi.org/10.5220/0011037900003209

Methods, Software, & Research Infrastructures

(2025) Developed a Gray–Scott simulation codebase for backpropagation-based parameter inversion: https://github.com/Yan-Yang-bot/bp2renderer, incorporating adaptive learning rates and embedded simulation animation. Repeated failure modes prompted codebase extensions for systematic loss landscape plotting, including visualization across parameter pairs and different loss functions. This leads to principled theoretical analyses and PINN design implications condensed into a submission to AI4Physics, ICML'26. *Independent Research @ self*

(2021 ~ 2023) Led prototyping of a CNN embedding visualization tool for cell images, , including installation guides for the local-running prototype; then redesigned the system for commercial lab product integration. My system design covered decisions on data storage media, database schemas, API specs, serialization/deserialization options, caching, and parallelizing large calculations, designed for incremental development and long-term scalability. *R&D Intern / R&D Engineer @ Deepcell Inc.*

(2018) Revamped search engine's VR interface (hand model calibration and connection) for behavioral data collection and provided web technique support for its SIGIR conference demonstration, work acknowledged by the corresponding author as authorship-worthy. *Research Assistant @ Georgetown University*

(2015) Led full-stack development of a weather monitoring application with animated charts. Implemented precipitation interpolation and data API monitoring. Technologies: AngularJS, Flask, NodeJS, AliCloud, Nginx, MySQL. *Full Stack Engineer @ ColorfulClouds Tech Co. Ltd*

Scientific and Societal Impact

(Oct 2026, upcoming) Poster Presenter @ Models of Consciousness 7, AMCS, University of Copenhagen
Yang, Y. Toward a Formal Account of Reversal Triggers: From Allostatic Regulation to Compositional-Distributional Semantics in Reversal Theory

(July 2026) Poster Presenter @ AI4Physics workshop, ICML 2026
Yang, Y. *Loss Landscape Diagnosis for Gradient-Based Gray-Scott System Inversion: Disentangling the Roles of PINN Components*

(Sep – Oct 2025) Active Participant @ Models of Consciousness Conference 6
Introduced Gödel's incompleteness theorem into meta-discussions on consciousness modeling.

(Apr 2025 – Mar 2026) LinkedIn creator (content relocated to SubStack: https://substack.com/@yy490)
Visualizations and essays for mathematical understanding:
- *Animation: Dirac delta approximation via cosine summation*
- *Visual diagram of matrix classifications in advanced linear algebra*
- *Short essay on the development of non-Euclidean geometry, with a visual timeline*

(Sep 2022) Active Participant @ Models of Consciousness Conference 3
Raised discussion on temporal scales of psychological processes (short-term fluctuations vs long-term life-shaping trajectories) and related data collection; engaged in post-talk exchanges with faculty from CMU (CS, subconscious models) and Stanford (philosophy of mind, qualia).

(Apr 2022) Presenter @ IMPROVE (online due to COVID-19)
Yang, Y., Bebis, G., & Nicolescu, M. *Two-Step Data Augmentation for Masked Face Detection & Recognition*.

Language Skills

Mandarin – Native **English** –Fluent (six-year US residency; graduate study and eight years of scientific/technical work in English) **French** – Basic beginner proficiency (self-assessed) **Traditional Chinese** – Basic elementary proficiency (self-assessed)

Chronological Work Experience[1]

Jul – Nov 2025 Reviewer @ IEEE Transactions on Affective Computing

[1] *Intentionally arranged chronologically to reflect the evolution of my research and professional interests*

Reviewed EEG-related ML manuscripts: corrected two serious conceptual misuses of cross-disciplinary terminology in a manuscript; identified deeper questions for empirical interpretation; and more.

Oct 2025 – Current Independent Researcher

Independently conceived, resourced, and led a self-directed research agenda across two active lines of work (physics-informed ML and motivational-state theory) without institutional affiliation or funding, from problem formulation through publication.

• Diagnosed gradient-based inversion for the Gray-Scott reaction-diffusion system by backpropagating through an unrolled simulation and used this as an ablation to analyze PINN components — presented at the AI4Physics workshop @ ICML as a poster, with code released on GitHub.

• Published a commentary re-examining a prior co-authored paper applying Reversal Theory (RT) to search-behavior modeling; identified mismatches between the theory's phenomenological commitments and the paper's statistical operationalizations, and distinguished theory-inspired approximations from faithful implementations — in the Journal of Motivation, Emotion, and Personality.

• Ongoing work formalizing RT — accepted as a poster at Models of Consciousness 7 (Copenhagen, Oct 2026) — with research notes on Zenodo extending RT with linkages to Barrett's predictive allostasis framework and to quantum measurement principles; one manuscript undergoing major revision for resubmission to TMLR.

• Participated in BAMΞ workshop on measurement (online), Models of Consciousness Conference 6 (Sapporo, Japan), and IBRO Autumn School on fMRI (NYU Abu Dhabi).

Nov 2024 – Sep 2025 <Career Gap> *Freelance work, independent writing, medical recovery, and other activities listed in this CV within the same time period.*

Jul 2024 – Jan 2025 Reviewer @ IEEE Transactions on Neural Networks and Learning Systems

Sep 2023 – Nov 2024 Ph.D. Research Fellow @ KAUST (Thuwal, Saudi Arabia)[2]

• Investigated learning of dynamical system's parameters from data, a mathematical framework with direct connections to neuroscience modeling, which motivated my pursuit of the direction; this project aims to apply it to automate synthesis of diseased leaf images for agricultural classification models.

• Proposed ideas on Arabic Speech Recognition, including acoustic source separation and data augmentation in the spectrogram.

May – Aug 2023 <Career Gap> *Personal transition period following a layoff; Ph.D. offer decision.*

Oct 2022 – May 2023 Machine Learning R&D Engineer @ Deepcell Inc. (Mountain View, United States)

Led system design for remodeling the Neural Network embedding visualization tool, making it the human-in-the-loop decision point within the commercial Cluster-Based Sorting product: scientists in customer labs select sorting criteria in an interactive exploration session before the run proceeds.

May 2021 – Sep 2022 R&D Intern @ Deepcell Inc. (Mountain View, United States)

• Trained GAN models to synthesize cell image data, augmenting training sets for downstream classifiers.

• Led development of a prototype CNN embedding visualization tool adopted by in-house data scientists and

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[2] *Throughout my time at KAUST and after my departure, I sustained independent study and self-directed research: a paper presented at AI4Physics@ICML 2026 and now under major revision for resubmission to TMLR, a commentary published in a psychology journal, a poster accepted to Models of Consciousness 7. Methodological and data integrity concerns I raised at KAUST were repeatedly dismissed, leading to my withdrawal from the program. Some of these concerns are under journal retraction review:* https://pubpeer.com/publications/7893CE90BA51FEEDCEF83F23B88FFA.

bioinformaticians for interactive clustering and correlation analysis; the tool became a direct precursor to the commercial Cluster-Based Sorting product built in the subsequent full-time role.

Jan – May 2022 Teaching Assistant @ University of Nevada, Reno (Reno, United States)

Recitation, grading, and office hours for MATH 127 (Precalculus II) and CS 320 (Data Structures); exploratory visit to the Visual Perception Lab surveying Representational Similarity Analysis, Multivoxel Pattern Analysis, and fMRI-based face/object perception, with training in BrainVoyager and human research ethics.

Sep 2021 – May 2022 Student Researcher @ University of Nevada, Reno (Reno, United States)

• Engaged in a variety of course research projects, such as HPC log data analytics, medical data segmentation and classification, and adversarial learning used for data augmentation.
• Led the research agenda for Two-Step Data Augmentation for Masked Face Detection and Recognition (IMPROVE 2022).

Jun 2020 – Jan 2021 Member of Technical Staff @ Pure Storage (Bellevue, United States)

• Implemented feature: import/restore volume snapshots from company disks into Kubernetes.
• Led the GitHub release process of Pure Service Orchestrator v 6.0.3.

Jun 2019 – May 2020 <Career Gap> *Coursework makeup after medical leave; job search delayed by COVID onset.*

Jan 2018 – Sep 2019 Research Assistant @ Georgetown University (Washington, United States)

• Led a critical step in anchoring the formalization of Reversal Theory's introspectively grounded motivational states—no prior math operationalization in the literature. The step grounded the formalization in RT's phenomenology and helped improve user struggle detection via search log data analysis. (InfoSense Lab)
• Revamped the search engine's VR interface for behavioral data collection, and provided web technique support for its SIGIR '18 conference demonstration, work acknowledged by the corresponding author as authorship-worthy. (InfoSense Lab)
• #MeToo Tweets Analytics (Georgetown course project, 3.4M tweets): applied LDA topic modeling (BoW & TF-IDF) to surface discourse themes (e.g., harassment/assault language, #MeToo historical framing, politicized subtopics); separately compiled GloVe-expanded occupational word-count analysis referencing U.S. Bureau of Labor Statistics categories (Massive Data Institute)

Sep 2016 – Jul 2017 <Career Gap> ***Language training*** *and graduate-school applications.*

Sep 2013 – Aug 2016 Software/Web Engineer & Mentor @ various companies (Beijing, China)

Transitioned from translation into software engineering via self-study in C++ (independently built a C++ minesweeper game as a first project), then pursued full-stack development across web apps, mobile mini-games, and desktop tools; mentored new employees on web development; stack included React-Redux, WebSocket, AngularJS, GoLang, Python, Redis database, NumPy, D3.js, C++.

Jul 2006 – Sep 2013 Scientific Secretary @ Tongji University (Shanghai, China) / Psychology Peer Supporter (Volunteer) @ Ayaohelp.com (Remote, China) / Translator (EN-CN) @ Self (Beijing, China)

• Provided peer support drawing on self-study in psychology; essays/poems on personal development were highlighted on front page by the site host, a Psychology professor at Beijing Normal University.
• Led a collaborative translation of Women in the 19th Century (Margaret Fuller, 1843) to completion, coordinating editorial review; translated/proofread marketing, technical, and media content, including science news for outlets such as Wired.com.